

Research Engineer at the Max Planck Institute for Intelligent Systems in Tübingen, working on 3D digital humans in the team of Michael J. Black. My research centers on parametric body models (SMPL-X, MANO, and FLAME) and their use in human-object interaction, video diffusion models, and face reconstruction. Day-to-day, I work with PyTorch, building data and rendering pipelines, then fine-tuning and evaluating models. I am interested in using LLMs as agents that automatically assemble, run, and evaluate computer-vision pipelines, turning a described business problem into a working vision solution.

References: [Michael J. Black](#) & [Victoria Fernández Abrevaya](#)

EXPERIENCE

Max Planck Institute for Intelligent Systems, Perceiving Systems

02/2024 – Present

Research Engineer

Tübingen, Germany

- First-authored InterDyn (CVPR 2025) on video diffusion models for object dynamics, and co-authored a second CVPR 2025 paper on occluded face expression reconstruction.
- Through shipping two CVPR papers end-to-end, learned to take research from idea to camera-ready: scoping experiments, building data and rendering pipelines, and iterating quickly under deadlines.
- Gave a spotlight talk at the Agents in Interaction: From Humans to Robots (H2R) workshop at CVPR 2025.
- Worked across the full AI stack for digital humans: LLMs and VLMs, video diffusion, and the SMPL family (SMPL-X, FLAME, MANO).
- Experienced across the full data and modelling loop: collecting, setting up, and processing datasets, and fine-tuning, evaluating, and reporting on models.

Loyens & Loeff

06/2023 – 08/2023

Working Student, Technology & Innovation

Amsterdam, The Netherlands

Supported the adoption of AI-powered legal-tech tools (including automated drafting) at a leading law firm, while researching the legal-AI market to advise on solution selection for specific business cases.

Innovation Centre for Artificial Intelligence

08/2022 – 03/2023

Communication Manager (self-employed)

Amsterdam, The Netherlands

Grew the public profile of UvA's AI innovation center. I led interview campaigns with AI researchers, helped organise national AI events, and produced and distributed all web, LinkedIn, and social content.

Perron038

02/2021 – 05/2021

Machine Learning Internship

Zwolle, The Netherlands

Prototyped a non-destructive method to detect leakage invisible from the outside of sealed detergent-pod containers using active thermography; co-designed a cobot-driven capture-and-heating rig for repeatable imaging and trained classifiers on around 1,000 labelled thermal images, demonstrating reliable leak detection.

Tembo Group

05/2021 – 08/2021

Design Science Internship

Kampen, The Netherlands

Investigated the quality trade-offs of cross-silo horizontal federated learning for industrial machinery manufacturers via a systematic literature review, then designed and validated a framework for navigating competing requirements (privacy, communication efficiency) through a UTAUT-based case study.

LIDL Netherlands Headquarters

07/2018 – 03/2019

Customer Service Representative

Huizen, The Netherlands

Handled 100+ requests per day by phone and email, resolving service issues at LIDL's national headquarters.

EDUCATION

University of Amsterdam, Master in Artificial Intelligence 09/2021 – 08/2024

- Research-centered program — 3.8/4.0 GPA
- Thesis: Synthesizing Object Dynamics from Human Motion with a Video Generative Prior (8/10) [↗](#)

University of Twente, Bachelor in Technology and Liberal Arts & Sciences 09/2018 – 08/2021

- Focused on machine learning and data science — 4.0/4.0 GPA
- Honours Programme: Processes of Change (additional 30 ECs)
- Thesis: Non-functional Requirements, Solutions and Interdependencies in Cross-Silo Horizontal Federated Learning

PUBLICATIONS

- **InterDyn: Controllable Interactive Dynamics with Video Diffusion Models** [Paper](#) — [Website](#)
R. Akkerman, H. Feng, M. J. Black, D. Tzionas, V. Fernández Abrevaya
in Computer Vision and Pattern Recognition (**CVPR**), 2025
- **OFER: Occluded Face Expression Reconstruction** [Paper](#) — [Website](#)
P. Selvaraju, V. Fernández Abrevaya, T. Bolkart, R. Akkerman, T. Ding, F. Amjadi, I. Zharkov
in Computer Vision and Pattern Recognition (**CVPR**), 2025
- **[Re] Explaining in Style: Training a GAN to explain a classifier in StyleSpace** [Paper](#)
N. Van der Vleuten, T. Radusinović, R. Akkerman, M. Reksoprodjo (equal contribution)
in ReScience C, 8(2), 2022; ML Reproducibility Challenge (**MLRC**) 2021

VOLUNTEERING

CVPR & Eurographics 2025 – 2026

Peer Reviewer

Reviewed conference submissions in computer graphics and computer vision.

University of Twente Innovation Fellows 10/2019 – 03/2024

Fellow / Coordinating Team

Enschede, The Netherlands

Directed a team of 12 students, overseeing strategic campus initiatives and co-managing an annual operating budget of €60,000; organised the 24-hour Twente-4-Hours event (KLM, Strategyzer, Stanford, FH Salzburg) and mentored new Fellows cohorts at TU Hamburg and Erasmus Rotterdam through Design Thinking training.

DIH-HERO 02/2021 – 08/2021

Grants & Contracts Support Officer

Enschede, The Netherlands

Supported coordination of a €16M Horizon 2020 action building a pan-European network of healthcare-robotics Digital Innovation Hubs across 17 partners, contributing to communications, events, and operations.

Autumn Challenge 08/2020 – 01/2021

Coach

Advised an international student team developing a Care Hotel concept report.

Stichting Huizer Bidders 01/2017 – 08/2018

Secretary, De Bramzigers

Huizen, The Netherlands

Co-established “Bramzigers”, the foundation’s youth wing, to pass on traditional botter craftsmanship and engage young members in preserving Huizen’s maritime heritage.

SKILLS

- **Technical:** Python, Bash, PyTorch, Hugging Face Transformers, PyTorch3D (differentiable rendering), PyRender/EGL (headless cluster rendering), NumPy, OpenCV, smplx & SMPL-family body models, project-aria; distributed multi-GPU training on A100/H100 clusters via HTCCondor
- **Project & People:** Project management, cross-functional team leadership, budget ownership, event organisation, technical & scientific communication
- **Languages:** English (proficient), Dutch (native), German (intermediate), Spanish (learning)